



Efficient Text Analysis: A BERT-Based Approach to Named Entity Recognition (NER) and Classification for Malayalam Language

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Abstract The onset of the present millennium has witnessed a remarkable surge in the use of Indian language in various forms of multilingual interpersonal communication, including news forums, webpages, email, and social media conversations. The extensive dissemination of mobile phones has ignited an apparently unceasing utilization of well-known micro-blogging platforms—Facebook, Twitter, and Whatsapp—to express thoughts, remarks, and impressions in the field of natural language processing (NLP). The primary objective of this research is to investigate Efficient Text Analysis using Bidirectional Encoder Representations from Transformers (BERT) Based for Named Entity Recognition and Classification in the Malayalam language. To conduct Named Entity Recognition (NER) in the Malayalam dialect, which is widely spoken in Kerala, India, this study employs both deep learning and the BERT approach. It involves customization of a pre-trained multilingual BERT (m-BERT) model to accurately capture the unique phonetic features of Malayalam. Lastly, the performance of the proposed approach is validated and compared with existing recent studies based on evaluation metrics such as accuracy (%), precision (%), recall (%), and f1-score (%). The results

show that the proposed approach attains the highest accuracy (97.10%), thereby outperforming the existing recent studies.

Keywords NLP · Information Extraction · NER · Deep Learning · BERT

1 Introduction

NER is an act of recognizing and categorizing entities—people, places, and organizations—that have specific names interior a given content [1, 2]. Within the reach of NLP, it is a basic pre-processing step that helps within the extraction of structure and important data from unstructured content input. Applications include data recovery, machine interpretation, and assumption investigation [3, 4]. Indeed whereas NER for dialects like English has progressed essentially, but NER for dialects with small assets, like Malayalam, is still required. Several deep learning strategies are applied to classify Malayalam dialects but they are equipped with many challenges [5]. The linguistic idiosyncrasies of Malayalam give specific difficulties for NER. The dialect incorporates an extraordinary degree of morphological assortment due to its complex character, in which words are produced by joining morphemes. The incorporation of highly complex shapes and unlimited word vocabulary makes challenging task for normal NER models to generalize effectively. It is mandatory to handle such issues for NER as it is fully dependent on its sentence specific-setting. It can be possible with the help of BERT (Bidirectional Encoder Representations from Transformers) which is a viable strategy for NER in dialects like Malayalam, where words might have a few implications based on their setting, since it can capture these relevant affiliations. BERT's capacity to learn from tremendous volumes of unlabelled information is another advantage [6].

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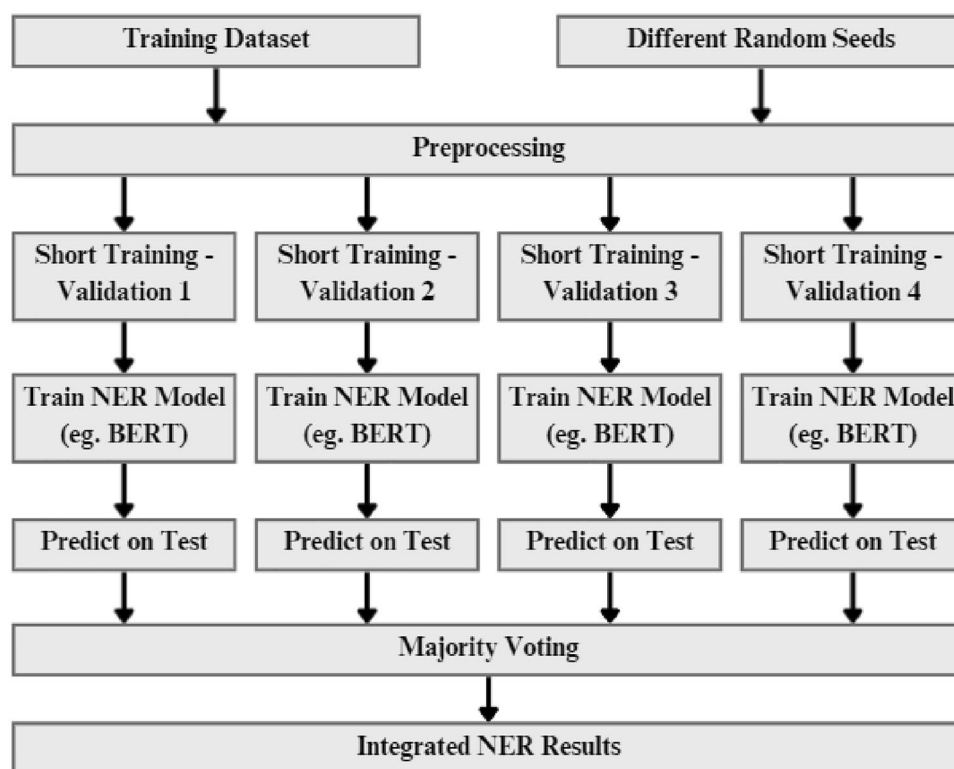
This capability may be particularly supportive for dialects like Malayalam, where BERT is able to alter to the special nuances and highlights of the dialect by being fine-tuned on a minor amount of labelled information. Therefore, BERT is the most suitable choice for NER in Malayalam and other dialects due to its flexibility in dealing with an assortment of NLP issues, as well as its capacity to recognize relevant joins and learn from endless volumes of information.

1.1 Novelty and contributions of the study

- Our proposed approach undergoes pre-training on a large corpus of text, utilizing masked language modelling and next sentence prediction. This process leads to comprehend word context within textual data and produce comprehensive word representations.
- The proposed model can be fine-tuned on a specific NER dataset based on hyperparameters such as learning rate, epochs, and batch size.
- Contextualized embedding for each token is generated in the form of word sequences, supplemented with prefix and suffix positions respectively.
- During encoding, each token representation is updated based on the contextual information from the surrounding tokens.
- The encoded sequence is then fed into a token classification layer to predict the label for each token in the input sequence. It leads to higher accuracy and efficiency of the proposed system.
- After generating results, the proposed system filters out labels with low confidence scores by merging adjacent tokens.

Table 1 Existing recent studies and their findings

Cited Works	Description/ Models used	Findings / Gaps identified
[7–9]	Named Entity Recognition (NER) and Classification for linguistics	In [7], the study tested three deep learning models for three shared task subtasks. Model-1 combines CNN, Bi-LSTM, and a fully connected layer in addition to the output layer. Model-2 has a Bi-LSTM layer, while Model-3 has a Bi-RNN layer. The study examined the training performance of Malayalam-English, Tamil-English, and Kannada-English datasets. The study presented Model-1's predictions based on training performance The results from study [8] are on European languages using linguistic features. Precision and recall in English are 84% and 80%, respectively. Spanish has an 85% f-score. Dutch achieved 90.3% precision and 92.23% recall. Hungarian earned an f-measure of 84.57%. German achieved 81% precision, 72% recall, and 76% f-measure In [9], the authors evaluated gaps in NER systems for Indian languages, particularly Hindi. Existing NER systems measure accuracy using predetermined datasets, resulting in inconsistent outcomes across datasets
[10]	Stacked autoencoder (AE) model	The experiments demonstrated that the system attains an accuracy of 95.3% tested for various network characteristics
[11]	Convolutional neural network (CNN) + Support vector machine (SVM)	Failed to predict label for each token in input sequence and attained an average accuracy of 76.23%
[12]	Long-short term memory (LSTM) + Naïve Bayes (NB)	The number of layers needed to design LSTM is unable to perform tokenization and achieved an accuracy of 84.56%
[13]	Bi-LSTM	Only performs the task of data pre-processing rather than generating word sequences and encoding them in dialect format
[14]	Gated recurrent unit (GRU)	Prone to over-fitting while performing tokenization of word sequences
[15]	Bi-GRU	Evaluated gaps in NER systems for Indian languages, particularly Hindi and Marathi, resulting in an accuracy of 89.12%
[16]	CNN + NB	Failed to predict label for each token in input sequence and attained an average accuracy of 73.53%
[17]	Recurrent neural network (RNN) + Decision tree (DT)	Encoding is somewhat difficult while traversing from leaf nodes to root node of generated lexicon tree
[18]	Deep belief networks (DBN's)	High computational costs and lack of accuracy in validating the system

Fig. 1 Workflow of the proposed framework

1.2 Organization of the given study

The given paper is categorized into the following sections. Section 2 presents an overview of existing studies in the context of NLP and classification of language. Section 3 describes the phases involved in the proposed system. Section 4 shows results and discussions. Section 5 concludes the given paper followed by references and future scope.

2 Literature Review

Table 1 shows an overview of existing studies and their findings in the context of NER [7–18].

3 Proposed Framework

Figure 1 presents workflow of an ensemble framework for moving towards NER and classification for Malayalam language utilizing a BERT-based approach [6, 11, 13, 14].

The process begins with a training dataset and different random seeds to introduce variability. Pre-processing is applied to the data, followed by short training-validation cycles using four different steps as shown below:

3.1 Inline Figure

Figure 2 shows a snippet performing pre-processing of data, tokenization of data, splitting of data and many more. The proposed framework is implemented in Python.

4 Results and Discussions

The given section presents generalization capabilities showing training and validation loss progression of a few existing deep learning models used in the context of NER in classification of Malayalam language. These plots are generally accomplished by means of the fine-tuning method. This study tailored a pre-trained multilingual BERT model to demonstrate special phonetic characteristics of Malayalam language by taking corpus into consideration. The method utilizes rich contextual information within the BERT design to comprehend the particular dialect designs of the Malayalam language. Figure 3 (a-f) shows the losses over epochs of existing deep learning models such as LSTM, GRU, Bi-GRU, Bi-LSTM, RNN and Bi-GRU. These graphs depict overfitting and under fitting when tested with existing models.

This section also compares the proposed framework with existing works in terms of results. Table 2 displays

```

python
1 import torch
2 from transformers import BertTokenizer, BertForTokenClassification, AdamW
3 from torch.utils.data import DataLoader, RandomSampler, SequentialSampler, TensorDataset
4 from sklearn.model_selection import train_test_split
5 from sklearn.metrics import classification_report
6
7 # Step 1: Data Preparation
8 def preprocess_data(texts, labels, tokenizer, max_len):
9     input_ids = []
10    attention_masks = []
11    label_ids = []
12
13    for text, label in zip(texts, labels):
14        encoded_dict = tokenizer.encode_plus(
15            text,
16            add_special_tokens=True,
17            max_length=max_len,
18            pad_to_max_length=True,
19            return_attention_mask=True,
20            return_tensors='pt',
21        )
22        input_ids.append(encoded_dict['input_ids'])
23        attention_masks.append(encoded_dict['attention_mask'])
24        label_ids.append(label)
25
26    return torch.cat(input_ids, dim=0), torch.cat(attention_masks, dim=0), torch.tensor(l
27
28 # Load your dataset
29 texts = ["മലയാളം വാക്കുകൾ", "മറ്റൊരു ഉദാഹരണം"]
30 labels = [[0, 1, 0], [0, 1, 1]] # Example labels
31
32 # Initialize tokenizer
33 tokenizer = BertTokenizer.from_pretrained('bert-base-multilingual-cased')
34
35 # Preprocess data
36 input_ids, attention_masks, label_ids = preprocess_data(texts, labels, tokenizer, max_len
37
38 # Split data
39 train_inputs, validation_inputs, train_labels, validation_labels = train_test_split(input
40 train_masks, validation_masks, __, __ = train_test_split(attention_masks, attention_masks,
41
42 # Create DataLoader
43 train_data = TensorDataset(train_inputs, train_masks, train_labels)
44 train_sampler = RandomSampler(train_data)
45 train_dataloader = DataLoader(train_data, sampler=train_sampler, batch_size=32)
46
47 validation_data = TensorDataset(validation_inputs, validation_masks, validation_labels)
48 validation_sampler = SequentialSampler(validation_data)
49 validation_dataloader = DataLoader(validation_data, sampler=validation_sampler, batch_siz
50
51 # Step 2: Model Setup
52 model = BertForTokenClassification.from_pretrained('bert-base-multilingual-cased', num_la
53
54 # Step 3: Training
55 optimizer = AdamW(model.parameters(), lr=2e-5)
56
57 for epoch in range(3):
58     model.train()
59     for batch in train_dataloader:
60         b_input_ids, b_input_mask, b_labels = batch
61         model.zero_grad()
62         outputs = model(b_input_ids, attention_mask=b_input_mask, labels=b_labels)
63         loss = outputs.loss
64         loss.backward()
65         optimizer.step()
66
67 # Step 4: Evaluation
68 model.eval()
69 predictions, true_labels = [], []
70
71 for batch in validation_dataloader:
72     b_input_ids, b_input_mask, b_labels = batch
73     with torch.no_grad():
74         outputs = model(b_input_ids, attention_mask=b_input_mask)
75     logits = outputs.logits
76     predictions.append(logits)
77     true_labels.append(b_labels)
78

```

Fig. 2 Code for Implementation

the comparison between the proposed work and previously published studies.

The comparison indicates that the proposed framework outperforms other techniques used in the current state of the art in terms of accuracy, precision, f1- score and recall.

The comparison is shown in Fig. 4. The proposed framework produces the highest precision (**95.37%**), highest recall (**96.55%**), highest f1- score (**96.45%**) and highest accuracy (**97.10%**).

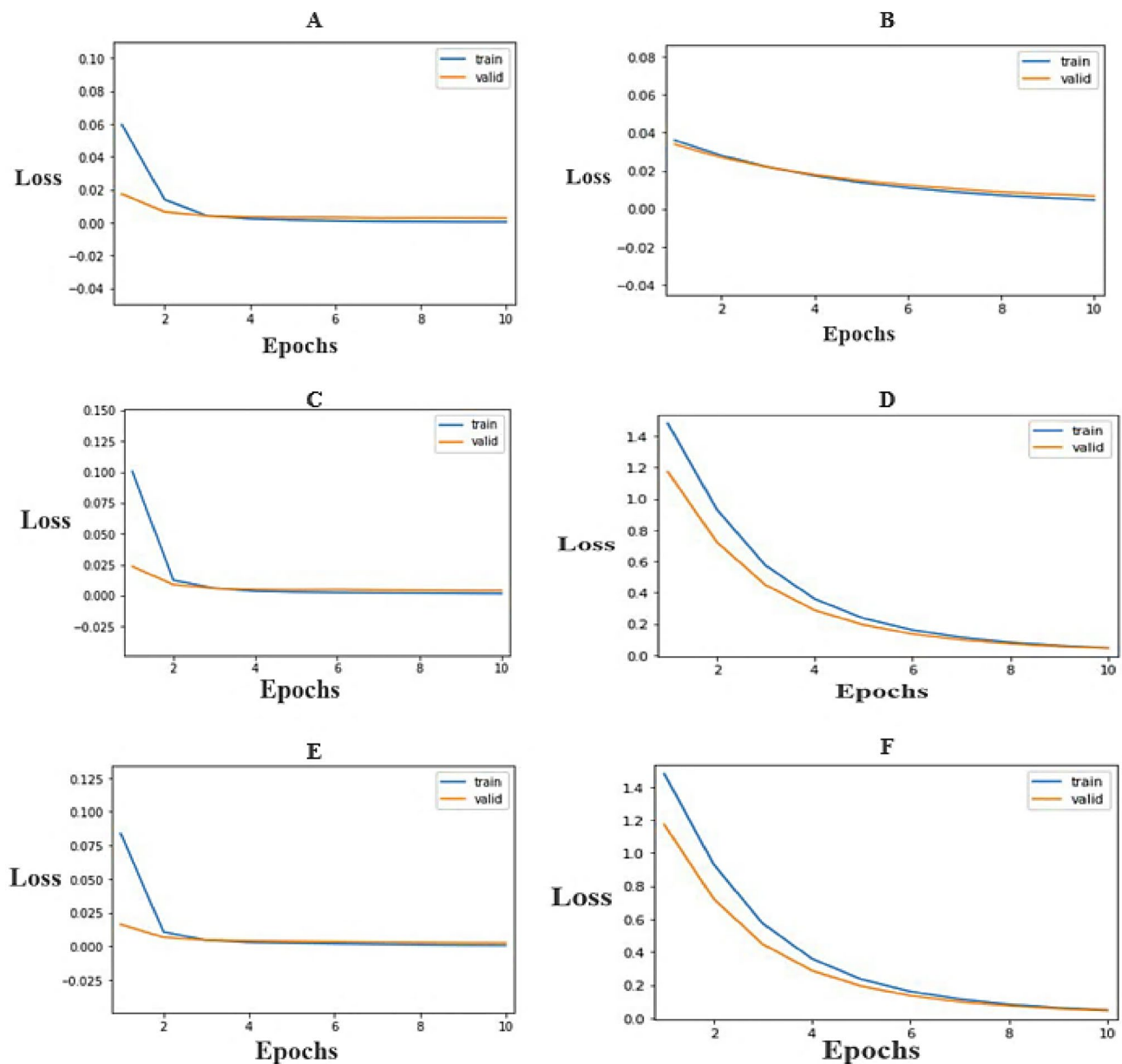


Fig. 3 Training and validation losses for (a) RNN (b) Bi-RNN (c) LSTM (d) Bi-LSTM (e) GRU (f) Bi-GRU

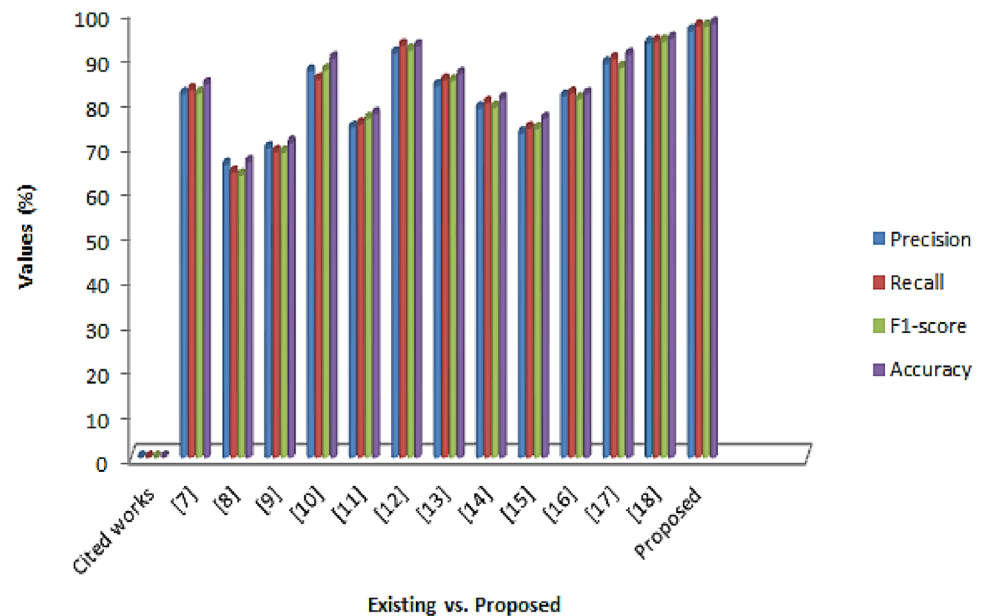
5 Conclusion and Future Scope

The given paper introduces an ensemble framework for NER and classification of Malayalam language using the m-BERT approach. The proposed framework makes use of a fine-tuned, pre-trained multilingual BERT approach on a clarified dataset, taking into consideration the specific etymological challenges displayed by Malayalam, such as its complex morphology. Text analysis involves pre-processing of the dataset, tokenization, normalization, and stemming of superfluous words from the corpus. It is followed by loading the model and adding an NER layer

to predict named entity tags. Then, the proposed model is trained and validated to tune hyperparameters and prevent overfitting / under fitting. Lastly, the model is validated with existing studies based on evaluation metrics such as accuracy, precision, recall, and f1-score. Our model attains the highest accuracy (97.10%), thus outperforming conventional deep learning models. Therefore, this extensive research on a BERT-based method for Named Entity Recognition (NER) and classification in Malayalam signifies notable progress in Natural Language Processing (NLP) for languages with limited resources.

Table 2 Validation of the proposed approach

Cited works	Precision (%)	Recall (%)	F1-Score (%)	Accuracy (%)
[7]	81.34	82.23	81.44	83.55
[8]	65.53	63.77	63.12	66.11
[9]	69.22	68.34	68.22	70.48
[10]	86.44	84.55	86.67	89.32
[11]	73.78	74.55	75.77	76.89
[12]	90.44	92.22	91.15	92.11
[13]	83.12	84.44	84.25	85.77
[14]	78.12	79.34	78.33	80.23
[15]	72.49	73.66	73.48	75.78
[16]	80.77	81.44	80.22	81.34
[17]	88.19	89.15	87.26	90.18
[18]	92.66	93.11	93.10	93.78
Proposed	95.37	96.55	96.45	97.10

Fig. 4 Comparison of the proposed approach with existing works based on metrics

Future research should prioritize the expansion of annotated datasets and the inclusion of varied Malayalam text sources to strengthen the model's resilience and investigate the potential for transfer learning from related languages. This will ultimately boost the accuracy and application of named entity recognition (NER) across different domains. Future research should focus on expanding the size of the annotated dataset to optimize the BERT model. The model's robustness and generalizability may be further improved by adding several sources of Malayalam text, such as literature, legal papers, medical data, and social media.

6 Conflict of Interest

The authors declare that there are no competing interests associated with this study.

References

- George L, Sumathy P (2023) An integrated clustering and BERT framework for improved topic modeling. *Int j inf tecnol* 15:2187–2195. <https://doi.org/10.1007/s41870-023-01268-w>
- Bajaj D, Bharti U, Gupta I et al (2024) GTMicro—microservice identification approach based on deep NLP transformer model

- for greenfield developments. *Int j inf tecnol* 16:2751–2761. <https://doi.org/10.1007/s41870-024-01766-5>
3. Mandava M, Vinta SR (2024) Optimized BERT: an effective attention layer based deep learning technique utilizing for multiword term extraction. *Int j inf tecnol*. <https://doi.org/10.1007/s41870-024-01855-5>
 4. Muralitharan J, Arumugam C (2024) Privacy BERT-LSTM: a novel NLP algorithm for sensitive information detection in textual documents. *Neural Comput & Applic* 36:15439–15454. <https://doi.org/10.1007/s00521-024-09707-w>
 5. Rajan R, Joseph AJ, Robin EK, Fathima NTK, "Part-Of-Speech Tagger in Malayalam Using Bi-directional LSTM," (2020) 23rd Conference of the Oriental COCOSDA International Committee for the Co-ordination and Standardisation of Speech Databases and Assessment Techniques (O-COCOSDA). Yangon, Myanmar 2020:22–27. <https://doi.org/10.1109/O-COCOSDA50338.2020.9295018>
 6. Sanjanasri JP, Vijay Krishna Menon S. Rajendran KP. Soman, and M. Anand Kumar. "Intrinsic evaluation for English–Tamil bilingual word embeddings." In *Intelligent Systems, Technologies and Applications: Proceedings of ISTA 2018*, pp. 39–51. Springer Singapore, 2020.
 7. Gupta I, Chatterjee I, Gupta N (2023) A two-staged NLP-based framework for assessing the sentiments on Indian supreme court judgments. *Int j inf tecnol* 15:2273–2282. <https://doi.org/10.1007/s41870-023-01273-z>
 8. M. Pathak and A. Jain, "µBoost: An Effective Method for Solving Indic Multilingual Text Classification Problem," *2022 IEEE Eighth International Conference on Multimedia Big Data (BigMM)*, Naples, Italy, 2022, pp. 96–100, <https://doi.org/10.1109/BigMM55396.2022.00022>.
 9. Rajan RP, Jose DV (2023) Text summarization using residual-based temporal attention convolutional neural network. *Int j inf tecnol*. <https://doi.org/10.1007/s41870-023-01581-4>
 10. Ma K, Tian M, Tan Y et al (2023) Ontology-Based BERT Model for Automated Information Extraction from Geological Hazard Reports. *J Earth Sci* 34:1390–1405. <https://doi.org/10.1007/s12583-022-1724-z>
 11. Menon Sreelekshmi JP. Sanjanasri B. Premjith and KP. Soman. "Named Entity Recognition Using Deep Learning and BERT for Tamil and Hindi Languages." *International Conference on Advances in Data Science and Computing Technologies*, pp. 395–403. Singapore: Springer Nature Singapore, 2022.
 12. Kumar SMU, Azim M, Quadri SMK (2024) Addressing the data gap: building a parallel corpus for Kashmiri language. *Int j inf tecnol* 16:4363–4379. <https://doi.org/10.1007/s41870-024-01979-8>
 13. Sreelakshmi K, Premjith B, and Soman Kp. "Amrita_CEN_NLP@ DravidianLangTech-EACL2021: deep learning-based offensive language identification in Malayalam, Tamil and Kannada." *Proceedings of the First Workshop on Speech and Language Technologies for Dravidian Languages*. 2021
 14. Pavan Kumar PHV, Premjith B, Sanjanasri JP, Soman KP. "Deep Learning Based Sentiment Analysis for Malayalam, Tamil and Kannada Languages." (2021).
 15. Chang, HY., Mozafari, S.H., Chen, C. *et al*. PipeBERT: High-throughput BERT Inference for ARM Big.LITTLE Multi-core Processors. *J Sign Process Syst* **95**, 877–894 (2023). <https://doi.org/10.1007/s11265-022-01814-y>
 16. N. Khamphakdee and P. Seresangtakul, "A Framework for Constructing Thai Sentiment Corpus using the Cosine Similarity Technique," *2021 13th International Conference on Knowledge and Smart Technology (KST)*, Bangsaen, Chonburi, Thailand, 2021, pp. 202–207, <https://doi.org/10.1109/KST51265.2021.9415802>.
 17. Kim Y, Kim JH, Lee JM et al (2022) A pre-trained BERT for Korean medical natural language processing. *Sci Rep* 12:13847. <https://doi.org/10.1038/s41598-022-17806-8>
 18. K. Sarveswaran and G. Dias, "Building a Part of Speech tagger for the Tamil Language," *2021 International Conference on Asian Language Processing (IALP)*, Singapore, Singapore, 2021, pp. 286–291, <https://doi.org/10.1109/IALP54817.2021.9675195>.

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